CLOSED-FORM GAUSSIAN INFORMATION PROJECTION FOR EMBEDDED SIMPLICES AND SIMPLICIAL MIXTURES

ALI DARIJANI[®□#¶⊥], EMADEDIN SOUFIAN[□#¶⊥], AND JÜRGEN BEYERER[®⊥]

Abstract. We study Gaussian approximation of Dirichlet–Gaussian probability density functions on embedded simplices. We derive the unique information projection onto the family of nondegenerate ambient Gaussian probability density functions, obtaining closed-form expressions for the optimal Gaussian and the minimum divergence for arbitrary positive Dirichlet parameters. The minimum divergence depends only on the intrinsic simplex dimension and Dirichlet parameters, is invariant up to an affine isomorphism, independent of the ambient dimension, and independent of the normal covariance. Furthermore, by joint convexity of Kullback–Leibler divergence, we establish explicit intrinsic-dimension-dependent sharp upper bounds for finite Gaussian Mixture with shared component weights. For uniform probability density functions, these bounds reduce to dimension-dependent constants. The results provide closed-form Gaussian Mixture constructions and rigorous approximation guarantees for probability density functions associated with embedded simplicial complexes. A numerical example on a circular curve illustrates the theoretical results and their connections to geometric Gaussian Mixture representations.

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Contents

1. Introduction 1
2. Previous Work 2
3. Construction 2
4. An Example 19
5. Conclusion 25
6. Acknowledgment 25
References 25
Authors' Affiliations 27

1. Introduction

Gaussian projection provides a canonical approximation of a *Probability Density Function* (PDF) by a Gaussian PDF [1, 2], minimizing the *Kullback–Leibler* (KL) divergence. For PDF associated with geometric simplicial structures [3], this problem admits additional structure through barycentric coordinates and the geometry of the embedding [4, 5, 6].

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We consider PDFs on embedded simplices generated by Dirichlet PDFs on the simplex tangent space and independent Gaussian PDFs on the simplex normal space. This formulation handles uniform PDF as well as non-uniform PDFs on simplices embedded in arbitrary finite ambient dimensions.

We derive the unique information projection [2] onto the Gaussian family, which is characterized by moment matching [1, 2], and obtain a closed-form expression for the information projection divergence. Remarkably, this quantity depends only on the intrinsic simplex dimension and the Dirichlet PDF parameters, and is invariant under affine isomorphism (by construction), independently of the ambient dimension and normal PDF a result similar to [6].

Furthermore, we extend the componentwise result to finite *Gaussian Mixture*s (GMs) of embedded simplicial PDFs. By joint convexity of the KL divergence [2], we establish explicit intrinsic-dimension-dependent sharp upper bounds for the corresponding GM. The uniform PDF follows as a special case. The main contributions are:

- Gaussian information projection: Characterization of the unique optimal Gaussian for embedded Dirichlet–Gaussian PDF.
- Closed-form divergence: Derivation of the information projection divergence and its invariance with respect to the affine isomorphism, ambient dimension, and normal PDF.
- GM approximation: Establishment of explicit information projection sharp upper bounds for finite GMs associated with embedded simplicial complex based PDFs.

The remainder of the paper develops the geometric and measure theoretic probability framework, derives the componentwise information projection, and establishes the corresponding information projection GM bounds.

2. Previous Work

This work generalizes our previous results on a GM representations of triangulated domains [6], which considered uniform PDF on full-dimensional simplices. We extend that framework in two directions:

- from uniform PDFs to arbitrary Dirichlet PDF with positive parameters, and
- from full-dimensional simplices to simplices embedded in arbitrary finite ambient dimensions with independent Gaussian PDF on their normal spaces.

These extensions give a general closed-form characterization of the componentwise Gaussian information projection and corresponding GM divergence bounds. The previous uniform model is recovered as a special case.

3. Construction

We first introduce the geometric and measure theoretic probability notation required to construct GM representation of embedded simplicial complexes. Throughout this paper, the ambient space is $\mathbb{R}^D$, whereas the simplices may have intrinsic dimensions $d \leq D$. Vectors are column vectors, I_D denotes the identity matrix on $\mathbb{R}^D$, and $A^\top$ denotes the transpose of a matrix A. For a symmetric positive-definite matrix Σ, $\mathcal{N}_D(x|m, \Sigma)$ denotes the Gaussian PDF on $\mathbb{R}^D$ with mean m and covariance Σ.

Preprint

Definition 3.1 (Convex Hull [3]). *Let $A \subseteq \mathbb{R}^D$. Its convex hull is defined by*

$$\operatorname{conv}(A) \coloneqq \left\{ \sum_{i=1}^{n} \lambda_i v_i : n \in \mathbb{N}_{>0}, v_i \in A, \lambda_i \geq 0, \sum_{i=1}^{n} \lambda_i = 1 \right\}. \tag{3.1}$$

Definition 3.2 (Standard Simplex [3]). *The standard d-simplex is*

$$\Delta^d \coloneqq \operatorname{conv}\{0, e_1, \ldots, e_d\} \subseteq \mathbb{R}^d, \tag{3.2}$$

where e_i denotes the i-th canonical basis vector. Equivalently,

$$\Delta^d = \left\{ x \in \mathbb{R}^d : x_i \geq 0, \quad \sum_{i=1}^{d} x_i \leq 1 \right\}. \tag{3.3}$$

Definition 3.3 (Geometrically Independent Points [3]). *$\{v_i\}_{i=0,\ldots,d} \subset \mathbb{R}^D$ are geometrically independent iff* $\operatorname{rank}[v_1 - v_0 \ldots v_d - v_0] = d$.

Definition 3.4 (Embedded Simplex [7]). *Let $v_0, \ldots, v_d \in \mathbb{R}^D$ be geometrically independent points; then*

$$\sigma^d = \operatorname{conv}\{v_0, \ldots, v_d\} \subseteq \mathbb{R}^D \tag{3.4}$$

is called an embedded d-simplex.

Remark 3.5. *The intrinsic dimension of an embedded simplex is d, independently of the ambient dimension D.*

Definition 3.6 (Faces of an Embedded Simplex). *Let*

$$\sigma^d = \operatorname{conv}\{v_0, \ldots, v_d\} \tag{3.5}$$

be an embedded simplex. A nonempty face of σ^d is a simplex of the form

$$\tau = \operatorname{conv}\{v_i : i \in I\}, \qquad \varnothing \neq I \subseteq \{0, \ldots, d\}. \tag{3.6}$$

Its dimension is

$$\dim \tau = |I| - 1, \tag{3.7}$$

where $|I|$ is the cardinality of the set I. In particular, its 0-faces, 1-faces, and $(d-1)$-faces are called vertices, edges, and facets, respectively.

Definition 3.7 (Simplicial Complex in $\mathbb{R}^D$). *A finite geometric simplicial complex $\mathcal{S}$ embedded in $\mathbb{R}^D$ is a finite collection of nondegenerate simplices satisfying:*

(1) Every nonempty face of a simplex in $\mathcal{S}$ also belongs to $\mathcal{S}$.
(2) For any two simplices $\sigma, \tau \in \mathcal{S}$, their intersection is either empty or a common face of both simplices.

The geometric realization of $\mathcal{S}$ is

$$|\mathcal{S}| \coloneqq \bigcup_{\sigma \in \mathcal{S}} \sigma \subseteq \mathbb{R}^D. \tag{3.8}$$

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Definition 3.8 (Dimension and Purity)**.** *Let $\mathcal{S}$ be a finite simplicial complex. Its intrinsic dimension is*

$$\dim \mathcal{S} \coloneqq \max_{\sigma \in \mathcal{S}} \dim \sigma. \tag{3.9}$$

A simplex $\sigma \in \mathcal{S}$ is called maximal if it is not a proper face of any other simplex in $\mathcal{S}$. The complex is called pure of dimension d if every maximal simplex has dimension d. We denote the collection of its d-dimensional simplices by

$$\mathcal{S}^{[d]} \coloneqq \{\sigma \in \mathcal{S} : \dim \sigma = d\}. \tag{3.10}$$

Definition 3.9 (Embedded Triangulation)**.** *Let $\Omega \subseteq \mathbb{R}^D$ be a set with intrinsic dimension d and finite positive d-dimensional Hausdorff measure [8]. A finite embedded triangulation of Ω is a finite, pure, d-dimensional simplicial complex $\mathcal{S}$ such that $|\mathcal{S}| = \Omega$. We denote the intrinsic d-dimensional volume by Vol_d, which agrees with the d-dimensional Hausdorff measure on the affine hull of each simplex.*

Proposition 3.10 (Intersections of Maximal Simplices)**.** *Let $\mathcal{S}$ be a finite, pure, d-dimensional simplicial complex embedded in $\mathbb{R}^D$. For distinct $\sigma, \tau \in \mathcal{S}^{[d]}$,*

$$\sigma \cap \tau = \varnothing \quad \text{or} \quad \sigma \cap \tau \in \mathcal{S}, \quad \dim(\sigma \cap \tau) \leq d - 1. \tag{3.11}$$

Consequently,

$$\mathrm{Vol}_d(\sigma \cap \tau) = 0. \tag{3.12}$$

Proof. By Definition 3.7, the intersection of two simplices is either empty or a common face. Since σ and τ are distinct d-simplices, their common face has dimension at most $d-1$. Every lower-dimensional face has zero d-dimensional Hausdorff measure. □

Corollary 3.11 (Volume of an Embedded Triangulation)**.** *Let $\mathcal{S}$ be a finite, pure, d-dimensional simplicial complex embedded in $\mathbb{R}^D$. Then*

$$\mathrm{Vol}_d(|\mathcal{S}|) = \sum_{\sigma \in \mathcal{S}^{[d]}} \mathrm{Vol}_d(\sigma). \tag{3.13}$$

Proof. The geometric realization is the union of its maximal d-simplices. By Proposition 3.10, distinct maximal simplices intersect only in sets of zero d-dimensional volume. Finite additivity of Hausdorff measure therefore gives the result. □

Proposition 3.12 (Affine Parametrization of an Embedded Simplex)**.** *Let*

$$\sigma^d = \mathrm{conv}\{v_0, \ldots, v_d\} \subseteq \mathbb{R}^D \tag{3.14}$$

be an embedded simplex. Define its edge matrix

$$E_d \coloneqq \begin{bmatrix} v_1 - v_0 & \cdots & v_d - v_0 \end{bmatrix} \in \mathbb{R}^{D \times d}. \tag{3.15}$$

Then the unique affine bijection from the standard simplex is

$$F : \Delta^d \to \sigma^d, \quad x \mapsto E_d x + v_0. \tag{3.16}$$

Proof. Since the vertices are geometrically independent, E_d has full column rank. Therefore, F is injective. Moreover,

$$F(0) = v_0, \quad F(e_i) = v_i, \quad i = 1, \ldots, d. \tag{3.17}$$

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By affinity, F maps the convex hull of the vertices of Δ^d onto the convex hull of the vertices of σ^d. Hence F is an affine bijection. Uniqueness follows because an affine map is uniquely determined by its values on affinely independent points. □

Proposition 3.13 (Intrinsic Volume of an Embedded Simplex)**.** *Let $\sigma^d = \mathrm{conv}\{v_0, \dots, v_d\}$ be an embedded simplex in $\mathbb{R}^D$. Its intrinsic volume is*

$$\mathrm{Vol}_d(\sigma^d) = \frac{1}{d!}\sqrt{\det(G_d)}, \tag{3.18}$$

where

$$G_d \coloneqq E_d^\top E_d \in \mathbb{R}^{d\times d} \tag{3.19}$$

is the Gram matrix of its edge vectors. In particular, G_d is positive definite.

Proof. The affine parametrization in Proposition 3.12 has constant d-dimensional volume factor

$$J_d(F) = \sqrt{\det(E_d^\top E_d)}. \tag{3.20}$$

Since

$$\mathrm{Vol}_d(\Delta^d) = \frac{1}{d!}, \tag{3.21}$$

the change-of-variables formula for the affine map gives the stated result. □

Definition 3.14 (Barycentric Subdivision)**.** *Let $\sigma^d = \mathrm{conv}\{v_0, \dots, v_d\}$ be a simplex. For each nonempty subset $I \subseteq \{0, \dots, d\}$, define its barycenter by*

$$b_I \coloneqq \frac{1}{|I|}\sum_{i\in I} v_i. \tag{3.22}$$

The barycentric subdivision of σ^d consists of the simplices whose vertices are barycenters of nested sequences of nonempty faces:

$$b_{I_0}, b_{I_1}, \dots, b_{I_r}, \quad I_0 \subsetneq I_1 \subsetneq \cdots \subsetneq I_r. \tag{3.23}$$

For a d-simplex, its maximal subdividing simplices correspond to permutations $(i_0, \dots, i_d)$ of $\{0, \dots, d\}$ and have vertices

$$b_{\{i_0\}}, b_{\{i_0,i_1\}}, \dots, b_{\{i_0,\dots,i_d\}}. \tag{3.24}$$

These simplices form a triangulation of the original simplex.

Remark 3.15. *Applying barycentric subdivision consistently to all simplices of an embedded simplicial complex produces a finer embedded simplicial complex without changing its geometric realization. In particular, neighboring simplices induce identical subdivisions on their common faces. This construction allows local refinement of an embedded mesh. The GM representation can subsequently be applied to the refined simplices using their respective vertices, intrinsic dimensions, and intrinsic volumes.*

Definition 3.16 (Simplex Tangent and Normal Spaces)**.** *Let*

$$\sigma^d = \mathrm{conv}\{v_0, \dots, v_d\} \subseteq \mathbb{R}^D \tag{3.25}$$

be an embedded simplex with $d < D$. Its tangent space is

$$T_\sigma \coloneqq \mathrm{span}\{v_i - v_0 : i = 1, \dots, d\}. \tag{3.26}$$

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Its normal space is the orthogonal complement

$$N_\sigma := T_\sigma^\perp = \{z \in \mathbb{R}^D : z^\top t = 0 \text{ for every } t \in T_\sigma\}. \tag{3.27}$$

Consequently,

$$\mathbb{R}^D = T_\sigma \oplus N_\sigma, \quad \dim T_\sigma = d, \quad \dim N_\sigma = D - d. \tag{3.28}$$

The affine hull of the simplex is $v_0 + T_\sigma$.

Remark 3.17 (Covariance Operators on Tangent and Normal Spaces)**.** *Throughout the paper, covariance operators on the tangent and normal spaces are distinguished from their ambient matrix representations.*
Let $U_N \in \mathbb{R}^{D\times(D-d)}$ have orthonormal columns spanning N_σ. The normal covariance operator is denoted by $\Sigma_N : N_\sigma \to N_\sigma$ and is positive definite on N_σ. Its matrix representation S_N in the orthonormal basis U_N is the unique matrix satisfying

$$\Sigma_N(U_N z) = U_N S_N z \quad \text{for every } z \in \mathbb{R}^{D-d}, \quad S_N \in \mathbb{S}_{++}^{D-d}. \tag{3.29}$$

The corresponding ambient representation, obtained by extending Σ_N by zero on T_σ, is denoted by

$$\overline{\Sigma}_N := U_N S_N U_N^\top \in \mathbb{S}_+^D. \tag{3.30}$$

Thus, $\overline{\Sigma}_N$ is positive semidefinite on $\mathbb{R}^D$, positive definite on N_σ, and vanishes on T_σ. In particular, it is generally singular as an ambient $D \times D$ matrix.

Definition 3.18 (Intrinsic Differential Entropy)**.** *Let $A = v_0 + T_\sigma$ denote the affine hull of the embedded simplex σ^d. We write $\mathcal{H}^d|_A$ for the restriction of the d-dimensional Hausdorff measure to A, with the standard normalization that agrees with ordinary Euclidean volume on d-dimensional affine spaces.*

For an Random Variable *(RV) Y supported on A and absolutely continuous with respect to $\mathcal{H}^d|_A$, let f_Y denote its PDF with respect to this measure. Its differential entropy is defined as*

$$h_{T_\sigma}(Y) := -\int_A f_Y(x) \log f_Y(x) \mathrm{d}\mathcal{H}^d(x), \tag{3.31}$$

provided the integral is well-defined. Equivalently, if $U_T \in \mathbb{R}^{D\times d}$ has orthonormal columns spanning T_σ, then $h_{T_\sigma}(Y)$ equals the ordinary differential entropy of the coordinate representation $U_T^\top(Y - v_0)$ in $\mathbb{R}^d$. Throughout the paper, differential entropies of RV supported on d-dimensional affine subspaces are understood in this intrinsic sense. Differential entropies of absolutely continuous PDF on $\mathbb{R}^D$ are understood with respect to D-dimensional Lebesgue measure.

Definition 3.19 (Dirichlet PDF on the Standard Simplex)**.** *Let*

$$\Delta^d = \left\{ x \in \mathbb{R}^d : x_i \geq 0,\ \sum_{i=1}^d x_i \leq 1 \right\}. \tag{3.32}$$

For $x \in \Delta^d$, define the barycentric coordinates

$$x_0 := 1 - \sum_{i=1}^d x_i. \tag{3.33}$$

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Given

$$\alpha_\sharp = (\alpha_0, \dots, \alpha_d)^\top \in \mathbb{R}^{d+1}_{++}, \tag{3.34}$$

the Dirichlet PDF is

$$\mathcal{D}_{\Delta^d}(x|\alpha_\sharp) = \frac{1}{\mathcal{B}(\alpha_\sharp)} \prod_{i=0}^{d} x_i^{\alpha_i - 1}, \tag{3.35}$$

where

$$\mathcal{B}(\alpha_\sharp) = \frac{\prod_{i=0}^{d} \Gamma(\alpha_i)}{\Gamma(|\alpha_\sharp|_1)}. \tag{3.36}$$

Corollary 3.20 (Uniform PDF on the Standard Simplex)**.** *For*

$$1_{d+1} \coloneqq (1, \dots, 1)^\top \in \mathbb{R}^{d+1}_{++}, \tag{3.37}$$

the Dirichlet PDF with parameter $\alpha_\sharp = 1_{d+1}$ is uniform on Δ^d:

$$\mathcal{D}_{\Delta^d}(x|1_{d+1}) = \begin{cases} d!, & x \in \Delta^d, \\ 0, & x \notin \Delta^d. \end{cases} \tag{3.38}$$

Proposition 3.21 (Mean and Covariance of the Dirichlet PDF)**.** *Let $X = (X_1, \dots, X_d)^\top$ follow the Dirichlet PDF with parameter $\alpha_\sharp \in \mathbb{R}^{d+1}_{++}$. Define*

$$\alpha_\natural \coloneqq (\alpha_1, \dots, \alpha_d)^\top, \quad a \coloneqq |\alpha_\sharp|_1. \tag{3.39}$$

Then

$$\mathbb{E}(X) = \frac{\alpha_\natural}{a}, \quad \mathbb{V}(X) = \frac{a \operatorname{diag}(\alpha_\natural) - \alpha_\natural \alpha_\natural^\top}{a^2(a+1)}. \tag{3.40}$$

Definition 3.22 (Dirichlet PDF on an Embedded Simplex)**.** *Let*

$$\sigma^d = \operatorname{conv}\{v_0, \dots, v_d\} \subseteq \mathbb{R}^D, \quad 1 \le d < D, \tag{3.41}$$

be an embedded simplex, and let $F(x) = E_d x + v_0$ be its affine parametrization. For $\alpha_\sharp \in \mathbb{R}^{d+1}_{++}$, the Dirichlet PDF on σ^d is the pushforward of the Dirichlet PDF on Δ^d under F. Equivalently, if

$$X \sim \mathcal{D}_{\Delta^d}(x|\alpha_\sharp), \tag{3.42}$$

then

$$Y \coloneqq E_d X + v_0 \tag{3.43}$$

follows the Dirichlet PDF on σ^d. We denote this PDF by $\mathcal{D}_{\sigma^d}(y|\alpha_\sharp)$.

Proposition 3.23 (Mean and Covariance on an Embedded Simplex)**.** *Under Definition 3.22, the mean and covariance of Y are*

$$m_T = \mathbb{E}(Y) = v_0 + E_d \frac{\alpha_\natural}{a}, \quad \Sigma_T = \mathbb{V}(Y) = E_d \frac{a \operatorname{diag}(\alpha_\natural) - \alpha_\natural \alpha_\natural^\top}{a^2(a+1)} E_d^\top, \tag{3.44}$$

where $a = \|\alpha_\sharp\|_1$. Moreover, Σ_T is positive definite on T_σ and vanishes on N_σ.

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Proof. The result follows directly from the affine transformation $Y = E_d X + v_0$ and Proposition 3.21. Since E_d has full column rank and the Dirichlet covariance is positive definite in $\mathbb{R}^d$, the covariance is positive definite on the range of E_d, which is precisely T_σ. □

Definition 3.24 (Forward KL Divergence)**.** *Let P and Q be probability measures on $\mathbb{R}^D$ that are absolutely continuous with respect to Lebesgue measure. Let p and q denote their respective Lebesgue densities. Their forward KL divergence is defined by*

$$D_{\mathrm{KL}}(P : Q) \coloneqq \int_{\mathbb{R}^D} p(x) \log\left(\frac{p(x)}{q(x)}\right) \mathrm{d}x, \tag{3.45}$$

The integral is understood in the extended-real sense, with $0 \log(0/q) = 0$. If P is not absolutely continuous with respect to Q, then

$$D_{\mathrm{KL}}(P : Q) = +\infty. \tag{3.46}$$

Definition 3.25 (Information Projection onto Gaussians)**.** *Let p be a probability density and let $\mathcal{G}_D$ denote the family of nondegenerate Gaussian PDFs on $\mathbb{R}^D$. The information projection of p onto $\mathcal{G}_D$ is defined as*

$$\Pi^{\mathrm{I}}_{\mathcal{G}_D}(p) \coloneqq \arg\min_{q \in \mathcal{G}_D} D_{\mathrm{KL}}(P : Q). \tag{3.47}$$

A Gaussian PDF with the same mean and covariance as p is called its moment-matched Gaussian.

Definition 3.26 (Dirichlet–Gaussian PDF on an Embedded Simplex)**.** *Let*

$$\sigma^d = \mathrm{conv}\{v_0, \ldots, v_d\} \subseteq \mathbb{R}^D, \quad 1 \le d < D, \tag{3.48}$$

be an embedded d-dimensional simplex with vertices $v_0, \ldots, v_d \in \mathbb{R}^D$.

Let

$$\Lambda = (\Lambda_0, \ldots, \Lambda_d)^\top \sim \mathcal{D}(\lambda | \alpha_\sharp), \quad \alpha_\sharp \in \mathbb{R}^{d+1}_{++}, \tag{3.49}$$

and define the tangential component

$$X_T \coloneqq \sum_{i=0}^{d} \Lambda_i v_i. \tag{3.50}$$

Since $\Lambda_i \ge 0$ and $\sum_{i=0}^d \Lambda_i = 1$ almost surely, X_T is a random convex combination of the vertices and is therefore supported on σ^d. Independently, let

$$Z_N \sim \mathcal{N}_{D-d}(z_N | 0, S_N), \quad X_N \coloneqq \sum_{j=1}^{D-d} (Z_N)_j u_j, \tag{3.51}$$

where $u_1, \ldots, u_{D-d}$ form an orthonormal basis of the normal space N_σ, and S_N is a positive definite covariance matrix on $\mathbb{R}^{D-d}$. Equivalently, X_N is a centered Gaussian RV supported on N_σ. Its covariance operator is denoted by Σ_N, and its covariance matrix when regarded as an $\mathbb{R}^D$-valued RV is denoted by $\overline{\Sigma}_N$. The Dirichlet–Gaussian random vector is defined as

$$X \coloneqq X_T + X_N = \sum_{i=0}^{d} \Lambda_i v_i + \sum_{j=1}^{D-d} (Z_N)_j u_j. \tag{3.52}$$

Preprint

Lemma 3.27 (Finiteness of the Dirichlet-Gaussian Entropy)**.** *Under the assumptions of Definition 3.26, the Dirichlet–Gaussian RV $X = X_T + X_N$ has a PDF with respect to D-dimensional Lebesgue measure and finite differential entropy.*

Proof. We first show that the Dirichlet PDF has finite differential entropy. For every $i \in \{0, \dots, d\}$, the marginal PDF of Λ_i is

$$\Lambda_i \sim \mathcal{B}(\lambda_i | \alpha_i, a - \alpha_i). \tag{3.53}$$

Since all Dirichlet parameters are strictly positive, both Beta parameters are positive. In particular,

$$\mathbb{E}(|\log \Lambda_i|) < +\infty. \tag{3.54}$$

Indeed, this follows from the integrability of $x^{\alpha_i - 1} |\log x|$ near zero and the analogous integrability near one. Using the Dirichlet log-density,

$$\log f_\Lambda(\Lambda) = -\log \mathcal{B}(\lambda | \alpha_\sharp) + \sum_{i=0}^{d} (\alpha_i - 1) \log \Lambda_i, \tag{3.55}$$

we obtain

$$\mathbb{E}(|\log f_\Lambda(\Lambda)|) < +\infty. \tag{3.56}$$

Consequently, the differential entropy of the Dirichlet PDF is finite. Since the embedding is nondegenerate, its Jacobian

$$J = \sqrt{\det(E_d^\top E_d)} \tag{3.57}$$

is strictly positive and finite. The affine change-of-variables formula therefore yields

$$h_{T_\sigma}(X_T) = h(\mathcal{D}(\lambda | \alpha_\sharp)) + \log J \in \mathbb{R}. \tag{3.58}$$

Now choose orthonormal basis matrices U_T and U_N of T_σ and N_σ, respectively, and write

$$X_N = U_N Z_N, \quad Z_N \sim \mathcal{N}_{D-d}(z_N | 0, S_N), \quad S_N \succ 0. \tag{3.59}$$

Consider the affine orthogonal coordinate transformation

$$\Phi(x) \coloneqq \begin{pmatrix} U_T^\top (x - v_0) \\ U_N^\top (x - v_0) \end{pmatrix}. \tag{3.60}$$

Since $[U_T \; U_N]$ is an orthogonal matrix, the absolute value of the Jacobian determinant of Φ is one. Under this transformation,

$$\Phi(X) = \begin{pmatrix} U_T^\top (X_T - v_0) \\ Z_N \end{pmatrix}, \tag{3.61}$$

because

$$X_T - v_0 \in T_\sigma, \quad U_N^\top (X_T - v_0) = 0, \tag{3.62}$$

and $X_N = U_N Z_N$. The two components are independent by construction. Consequently, the transformed RV has a product density, and entropy additivity gives

$$h(X) = h_{T_\sigma}(X_T) + h(Z_N) = h(\mathcal{D}(x_T | \alpha_\sharp)) + \log J + \frac{D - d}{2} \log(2\pi e) + \frac{1}{2} \log \det S_N. \tag{3.63}$$

Every term is finite. Thus X is absolutely continuous with respect to D-dimensional Lebesgue measure and has finite differential entropy. □

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Remark 3.28 (Nondegeneracy)**.** *The independent normal perturbation makes the resulting PDF absolutely continuous with respect to D-dimensional Lebesgue measure. This allows us to approximate the PDF by nondegenerate Gaussian PDFs in the ambient space.*

Proposition 3.29 (Gaussian Information Projection by Moment Matching)**.** *Let P be a probability distribution on $\mathbb{R}^D$ that is absolutely continuous with respect to Lebesgue measure, has finite differential entropy, finite first and second central moments, and positive-definite covariance matrix*

$$\Sigma = \mathbb{V}_P(X) \succ 0. \tag{3.64}$$

Let

$$m = \mathbb{E}_P(X). \tag{3.65}$$

Then the unique minimizer of the forward KL divergence over all nondegenerate Gaussian PDFs on $\mathbb{R}^D$ is

$$Q^* = \mathcal{N}_D(x|m, \Sigma). \tag{3.66}$$

The minimum divergence is

$$\inf_{\mu\in\mathbb{R}^D, S\succ 0} D_{\mathrm{KL}}\big(P : \mathcal{N}_D(x|\mu, S)\big) = \frac{1}{2}\log\left((2\pi e)^D \det\Sigma\right) - h(P). \tag{3.67}$$

Proof. For an arbitrary nondegenerate Gaussian distribution $Q_{\mu,S} = \mathcal{N}_D(x|\mu, S)$, the expected negative log-density under P gives

$$D_{\mathrm{KL}}(P : Q_{\mu,S}) = -h(P) + \frac{D}{2}\log(2\pi) + \frac{1}{2}\log\det S + \frac{1}{2}\operatorname{tr}(S^{-1}\Sigma) + \frac{1}{2}(m-\mu)^\top S^{-1}(m-\mu). \tag{3.68}$$

For fixed $S \succ 0$, the expression is uniquely minimized at $\mu = m$. Substituting $\mu = m$, it remains to minimize

$$\log\det S + \operatorname{tr}(S^{-1}\Sigma). \tag{3.69}$$

Set

$$A = \Sigma^{-1/2} S \Sigma^{-1/2}. \tag{3.70}$$

Then $A \succ 0$, and

$$\log\det S + \operatorname{tr}(S^{-1}\Sigma) = \log\det\Sigma + \log\det A + \operatorname{tr}(A^{-1}). \tag{3.71}$$

Let $\lambda_1, \ldots, \lambda_D > 0$ be the eigenvalues of A. The remaining terms become

$$\sum_{i=1}^{D}\left(\log\lambda_i + \frac{1}{\lambda_i}\right). \tag{3.72}$$

The scalar function

$$f(\lambda) = \log\lambda + \frac{1}{\lambda} \tag{3.73}$$

has derivative

$$f'(\lambda) = \frac{\lambda - 1}{\lambda^2}. \tag{3.74}$$

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It is strictly decreasing on $(0,1)$, strictly increasing on $(1,+\infty)$, and attains its unique global minimum $f(1)=1$. Thus $A=I_D$, equivalently $S=\Sigma$, is the unique minimizer. Substituting $\mu=m$ and $S=\Sigma$ into the divergence expression establishes both the claimed Gaussian projection and its minimum divergence. □

Theorem 3.30 (Gaussian Information Projection)**.** *Let p be the PDF of the Dirichlet–Gaussian RV in Definition 3.26. Let*

$$a := \|\alpha_\sharp\|_1, \quad \alpha_\natural := (\alpha_1,\dots,\alpha_d)^\top, \quad E_d = [v_1 - v_0, \dots, v_d - v_0]. \tag{3.75}$$

The unique information projection over all nondegenerate Gaussian PDFs on $\mathbb{R}^D$ is

$$q^* = \mathcal{N}_D(m_T, \Sigma_T + \overline{\Sigma}_N), \tag{3.76}$$

where

$$m_T = v_0 + E_d \frac{\alpha_\natural}{a}, \quad \Sigma_T = E_d \frac{a\,\mathrm{diag}(\alpha_\natural) - \alpha_\natural \alpha_\natural^\top}{a^2(a+1)} E_d^\top. \tag{3.77}$$

Furthermore

$$D_{\mathrm{KL}}(P : Q^*) = C_{d,\alpha_\sharp} \tag{3.78}$$

with

$$C_{d,\alpha_\sharp} = \frac{d}{2}\log(2\pi e) + \frac{1}{2}\left[\sum_{i=0}^{d} \log\alpha_i - (d+1)\log a - d\log(a+1)\right] - H(\alpha_\sharp), \tag{3.79}$$

where $H(\alpha_\sharp)$ is the differential entropy of the Dirichlet PDF on Δ^d:

$$H(\alpha_\sharp) = \log\left(\frac{\prod_{i=0}^{d}\Gamma(\alpha_i)}{\Gamma(a)}\right) + (a-d-1)\psi(a) - \sum_{i=0}^{d}(\alpha_i - 1)\psi(\alpha_i). \tag{3.80}$$

Here ψ denotes the digamma function. In particular, $C_{d,\alpha_\sharp}$ is independent of the ambient dimension, the embedding, the geometry and size of the simplex, and the normal PDF.

Proof. By Proposition 3.29, the information projection over nondegenerate Gaussian PDFs is the Gaussian with matching mean and covariance. By Lemma 3.27, the Dirichlet–Gaussian RV has finite differential entropy and is absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^D$. Furthermore, its tangent component is bounded because it is supported on the compact simplex σ^d, while its normal PDF is Gaussian. Hence X has finite first and second central moments. Since the tangent and normal covariance operators are positive definite on the complementary subspaces T_σ and N_σ, respectively, their sum is positive definite on $\mathbb{R}^D$. Therefore, all assumptions of Proposition 3.29 are satisfied. By Proposition 3.29, the uique information projection onto the Gaussian family is the Gaussian with matching mean and covariance. Since X_T and X_N are independent,

$$\mathbb{E}(X) = m_T, \quad \mathbb{V}(X) = \Sigma_T + \overline{\Sigma}_N. \tag{3.81}$$

The covariance is positive definite on $\mathbb{R}^D$, because Σ_T is positive definite on T_σ and $\overline{\Sigma}_N$ is positive definite on N_σ. Therefore, the unique Gaussian information projection is

$$q^* = \mathcal{N}_D\left(x | m_T, \Sigma_T + \overline{\Sigma}_N\right). \tag{3.82}$$

It remains to calculate the minimum divergence. Since X_T and X_N are independent,

$$\mathbb{E}(X) = m_T, \quad \mathbb{V}(X) = \Sigma_T + \overline{\Sigma}_N. \tag{3.83}$$

Preprint

The latter covariance is positive definite on $\mathbb{R}^D$, because Σ_T is positive definite on T_σ and $\overline{\Sigma}_N$ is positive definite on N_σ. Choose orthonormal coordinates adapted to

$$\mathbb{R}^D = T_\sigma \oplus N_\sigma. \tag{3.84}$$

In these coordinates, the target distribution and its moment-matched Gaussian factorize into their tangent and normal parts:

$$P = P_T \otimes P_N, \quad Q^* = Q_T^* \otimes P_N. \tag{3.85}$$

By invariance and additivity of KL divergence [9],

$$D_{\mathrm{KL}}(P : Q^*) = D_{\mathrm{KL}}(P_T : Q_T^*). \tag{3.86}$$

Let $V_d = \mathrm{Vol}_d(\sigma^d)$. The affine transformation from Δ^d to σ^d has volume factor

$$\sqrt{\det(E_d^\top E_d)} = d! V_d. \tag{3.87}$$

Consequently, using the intrinsic differential entropy defined above and the affine change-of-variables formula, we have

$$h_{T_\sigma}(X_T) = h(\mathcal{D}(x_T|\alpha_\sharp)) + \log\sqrt{\det(E_d^\top E_d)} = h(\mathcal{D}(x_T|\alpha_\sharp)) + \log(d! V_d). \tag{3.88}$$

Here $h(\mathcal{D}(x_T|\alpha_\sharp))$ denotes differential entropy with respect to d-dimensional Lebesgue measure on the standard simplex in reduced barycentric coordinates. Since Q_T^* is the Gaussian with the same mean and covariance as P_T, its differential entropy is

$$h(Q_T^*) = \frac{d}{2}\log(2\pi e) + \frac{1}{2}\log \operatorname*{det}_{T_\sigma}(\Sigma_T). \tag{3.89}$$

Thus

$$D_{\mathrm{KL}}(P_T : Q_T^*) = \frac{d}{2}\log(2\pi e) + \frac{1}{2}\log \operatorname*{det}_{T_\sigma}(\Sigma_T) - H(\alpha_\sharp) - \log(d! V_d). \tag{3.90}$$

To evaluate the determinant, let $U_T \in \mathbb{R}^{D\times d}$ have orthonormal columns spanning T_σ, and define

$$E \coloneqq U_T^\top E_d \in \mathbb{R}^{d\times d}. \tag{3.91}$$

Since $E_d = U_T E$, the covariance operator

$$\Sigma_T = E_d \Sigma_\lambda E_d^\top \tag{3.92}$$

has the matrix representation

$$[\Sigma_T]_{U_T} = E \Sigma_\lambda E^\top \tag{3.93}$$

in the orthonormal basis U_T. Consequently,

$$\operatorname*{det}_{T_\sigma}(\Sigma_T) = \det(E)^2 \det(\Sigma_\lambda). \tag{3.94}$$

Here, $\det_{T_\sigma}(\Sigma_T)$ denotes the determinant of the matrix representation of the restriction of Σ_T to T_σ in an orthonormal basis of T_σ. Using the Dirichlet covariance formula [10] and the matrix determinant lemma [11], we have

$$\det(\Sigma_\lambda) = \frac{\prod_{i=0}^d \alpha_i}{a^{d+1}(a+1)^d}. \tag{3.95}$$

Preprint

Furthermore,

$$\det(E)^2 = \det(E_d^\top E_d) = (d!V_d)^2. \tag{3.96}$$

Combining these expressions yields

$$\det_{T_\sigma}(\Sigma_T) = (d!V_d)^2 \frac{\prod_{i=0}^d \alpha_i}{a^{d+1}(a+1)^d}. \tag{3.97}$$

Substituting into the KL expression, the terms containing V_d cancel, giving

$$D_{\mathrm{KL}}(P : Q^*) = C_{d,\alpha_\sharp}. \tag{3.98}$$

This also establishes the independence of the minimum divergence from the embedding and normal covariance. □

Corollary 3.31 (Uniform Distribution on an Embedded Simplex)**.** *Under the assumptions of Theorem 3.30, let*

$$\alpha_\sharp = 1_{d+1}. \tag{3.99}$$

Then X_T is uniformly distributed on σ^d. Its mean and covariance are

$$m_T = v_0 + E_d \frac{1_d}{d+1}, \quad \Sigma_T = E_d \frac{(d+1)I_d - 1_d 1_d^\top}{(d+1)^2(d+2)} E_d^\top. \tag{3.100}$$

The minimum KL divergence is

$$C_d = \log\left(\frac{(2\pi e)^{d/2} d!}{(d+1)^{(d+1)/2}(d+2)^{d/2}} \right). \tag{3.101}$$

In particular, this value depends only on the intrinsic dimension d.

Definition 3.32 (Gaussian Mixture)**.** *A* Gaussian Mixture *in $\mathbb{R}^D$ is a PDF of the form*

$$q(x) = \sum_{j=1}^M w_j \mathcal{N}_D(x | m_j, \Sigma_j), \tag{3.102}$$

where

$$w_j \geq 0, \quad \sum_{j=1}^M w_j = 1, \quad \Sigma_j \in \mathbb{S}_{++}^D. \tag{3.103}$$

It can be interpreted as a two-stage random experiment: first select a component according to the weights $\{w_j\}_{j=1}^M$, then sample from the selected Gaussian PDF.

Proposition 3.33 (Uniform Distribution as a Mixture on an Embedded Complex)**.** *Let $\mathcal{S}$ be a finite, pure, d-dimensional simplicial complex embedded in $\mathbb{R}^D$, with geometric realization $\Omega = |\mathcal{S}|$ and*

$$0 < \mathrm{Vol}_d(\Omega) < +\infty. \tag{3.104}$$

Then the uniform probability measure on Ω can be represented as

$$\mathcal{U}_\Omega = \sum_{\sigma \in \mathcal{S}^{[d]}} w_\sigma \mathcal{U}_\sigma, \tag{3.105}$$

Preprint

where

$$w_\sigma = \frac{\mathrm{Vol}_d(\sigma)}{\mathrm{Vol}_d(\Omega)}, \quad \mathcal{U}_A \coloneqq \frac{\mathcal{H}^d|_A}{\mathrm{Vol}_d(A)}. \tag{3.106}$$

Here $\mathcal{H}^d|_A$ denotes the restriction of d-dimensional Hausdorff measure to A. In particular,

$$w_\sigma > 0, \qquad \sum_{\sigma \in \mathcal{S}^{[d]}} w_\sigma = 1.$$

Proof. By Proposition 3.10, distinct maximal simplices intersect only in sets of zero d-dimensional Hausdorff measure. Hence their uniform measures can be combined without double-counting positive-volume regions. Furthermore, by Corollary 3.11,

$$\mathrm{Vol}_d(\Omega) = \sum_{\sigma \in \mathcal{S}^{[d]}} \mathrm{Vol}_d(\sigma). \tag{3.107}$$

Therefore, for every measurable set $A \subseteq \Omega$,

$$\mathcal{U}_\Omega(A) = \frac{\mathcal{H}^d(A \cap \Omega)}{\mathrm{Vol}_d(\Omega)} = \sum_{\sigma \in \mathcal{S}^{[d]}} \frac{\mathrm{Vol}_d(\sigma)}{\mathrm{Vol}_d(\Omega)} \mathcal{U}_\sigma(A). \tag{3.108}$$

This proves the result. □

Proposition 3.34 (Joint Convexity of the KL Divergence [2])**.** *Let $p_1, \dots, p_M$ and $q_1, \dots, q_M$ be PDFs with respect to a common reference measure, and let*

$$\lambda_j \geq 0, \quad \sum_{j=1}^{M} \lambda_j = 1. \tag{3.109}$$

Then

$$D_{\mathrm{KL}}\left(\sum_{j=1}^{M} \lambda_j p_j : \sum_{j=1}^{M} \lambda_j q_j \right) \leq \sum_{j=1}^{M} \lambda_j D_{\mathrm{KL}}(p_j : q_j). \tag{3.110}$$

The inequality holds in the extended real sense.

Theorem 3.35 (GM Representation for Embedded Dirichlet–Gaussian PDFs)**.** *Let $\mathcal{S}$ be a finite, pure, d-dimensional simplicial complex embedded in $\mathbb{R}^D$, where $1 \leq d < D$, and let*

$$\{\sigma_j^d\}_{j=1}^M = \mathcal{S}^{[d]} \tag{3.111}$$

denote its maximal simplices. For each $j = 1, \dots, M$, let

$$\sigma_j^d = \mathrm{conv}\{v_{j,0}, v_{j,1}, \dots, v_{j,d}\} \tag{3.112}$$

be nondegenerate, and define the edge matrix

$$E_j = [v_{j,1} - v_{j,0}, \dots, v_{j,d} - v_{j,0}] \in \mathbb{R}^{D \times d}. \tag{3.113}$$

Let

$$\alpha_j = (\alpha_{j,0}, \dots, \alpha_{j,d}) \in (0, +\infty)^{d+1}, \quad a_j = \sum_{i=0}^{d} \alpha_{j,i}. \tag{3.114}$$

Let

$$X_j \sim \mathcal{D}(x_j | \alpha_j) \tag{3.115}$$

Preprint

denote the vector of barycentric coordinates, and define

$$Y_j = v_{j,0} + E_j \begin{pmatrix} X_{j,1} \\ \vdots \\ X_{j,d} \end{pmatrix}. \tag{3.116}$$

Independently, let

$$Z_j \sim \mathcal{N}_{N_j}(z_j|0, \Sigma_{N_j}), \tag{3.117}$$

where $N_j \coloneqq \mathrm{range}(E_j)^\perp$ *and* $\Sigma_{N,j} : N_j \to N_j$ *is positive definite. Write* $\overline{\Sigma}_{N,j} \in \mathbb{S}_+^D$ *for its ambient extension by zero on* $\mathrm{range}(E_j)$*. Thus, when* Z_j *is regarded as an* $\mathbb{R}^D$*-valued RV,* $\mathcal{V}(Z_j) = \overline{\Sigma}_{N,j}$*. Define the component distributions*

$$P_j = \mathcal{L}(Y_j + Z_j). \tag{3.118}$$

Let

$$\lambda_1, \dots, \lambda_M > 0, \quad \sum_{j=1}^M \lambda_j = 1, \tag{3.119}$$

be arbitrary mixture weights, and define

$$P = \sum_{j=1}^M \lambda_j P_j. \tag{3.120}$$

For each component, define

$$\alpha_{j,\natural} = (\alpha_{j,1}, \dots, \alpha_{j,d})^\top, \quad C_j = \frac{a_j \operatorname{diag}(\alpha_{j,\natural}) - \alpha_{j,\natural}\alpha_{j,\natural}^\top}{a_j^2(a_j+1)}. \tag{3.121}$$

Using the ambient extension $\overline{\Sigma}_{N,j}$ *defined above, define*

$$m_j = v_{j,0} + E_j \frac{\alpha_{j,\natural}}{a_j}, \quad \Sigma_j = E_j C_j E_j^\top + \overline{\Sigma}_{N,j}. \tag{3.122}$$

Consider the Gaussian PDF and their mixture

$$q_j = \mathcal{N}_D(q_j|m_j, \Sigma_j), \quad Q = \sum_{j=1}^M \lambda_j q_j. \tag{3.123}$$

Then the following statements hold.

(1) For every j*,* q_j *is the unique Gaussian information projection of* P_j *onto the family of nondegenerate Gaussian distributions on* $\mathbb{R}^D$*.*

(2) The global mixture divergence satisfies

$$D_{\mathrm{KL}}(P : Q) \le \sum_{j=1}^M \lambda_j C_{d,\alpha_j}, \tag{3.124}$$

where

$$C_{d,\alpha_j} = \frac{d}{2}\log(2\pi e) + \frac{1}{2}\left[\sum_{i=0}^d \log \alpha_{j,i} - (d+1)\log a_j - d\log(a_j+1)\right] - h(\mathcal{D}(\alpha_j)), \tag{3.125}$$

Preprint

and $h(\mathcal{D}(x_T|\alpha_j))$ denotes the differential entropy of the Dirichlet PDF on the standard d-simplex. In particular, the bound is independent of the ambient dimension, the simplex embeddings, the simplex volumes, and the normal covariance matrices.

Proof. We first establish the component-wise Gaussian information projections. Fix $j \in \{1, \ldots, M\}$ and consider

$$W_j = Y_j + Z_j. \tag{3.126}$$

By the moment formulas for the Dirichlet PDF,

$$\mathbb{E}(X_{j,\natural}) = \frac{\alpha_{j,\natural}}{a_j}, \quad \mathbb{V}(X_{j,\natural}) = C_j. \tag{3.127}$$

Consequently,

$$\mathbb{E}(W_j) = v_{j,0} + E_j \frac{\alpha_{j,\natural}}{a_j} = m_j. \tag{3.128}$$

Since Y_j and Z_j are independent,

$$\mathbb{V}(W_j) = E_j C_j E_j^\top + \overline{\Sigma}_{N,j} = \Sigma_j. \tag{3.129}$$

The matrix C_j is positive definite because $\alpha_{j,i} > 0$ for all $i = 0, \ldots, d$. Since E_j has full column rank, $E_j C_j E_j^\top$ is positive definite on the tangent space. By assumption, $\overline{\Sigma}_{N,j}$ is positive definite on the orthogonal normal space and vanishes on the tangent space. The tangent and normal spaces are orthogonal complements of $\mathbb{R}^D$. Hence, $\Sigma_j \succ 0$. Thus P_j has finite differential entropy and a positive-definite covariance matrix. By Theorem 3.30, its unique Gaussian information projection is the Gaussian distribution with matching mean and covariance, namely q_j. Furthermore, the component-level result gives

$$D_{\mathrm{KL}}(P_j : q_j) = C_{d,\alpha_j}. \tag{3.130}$$

We now establish the global mixture bound. By construction,

$$P = \sum_{j=1}^{M} \lambda_j P_j, \quad Q = \sum_{j=1}^{M} \lambda_j q_j. \tag{3.131}$$

All component distributions admit PDFs with respect to Lebesgue measure on $\mathbb{R}^D$. Moreover, their Gaussian information projections are nondegenerate Gaussian distributions. Thus the component-wise information projections are finite. Applying Proposition 3.34 to the component densities with their common mixture weights yields:

$$D_{\mathrm{KL}}(P : Q) = D_{\mathrm{KL}}\left(\sum_{j=1}^{M} \lambda_j P_j : \sum_{j=1}^{M} \lambda_j q_j\right) \leq \sum_{j=1}^{M} \lambda_j D_{\mathrm{KL}}(P_j : q_j). \tag{3.132}$$

Substituting the component-wise divergence expressions gives

$$D_{\mathrm{KL}}(P : Q) \leq \sum_{j=1}^{M} \lambda_j C_{d,\alpha_j}. \tag{3.133}$$

This proves the claimed inequality. Finally, each component constant depends only on the intrinsic dimension and its Dirichlet PDF parameters, by Theorem 3.30. Therefore, the global upper bound is independent of the ambient dimension, simplex embeddings, simplex

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volumes, and normal covariance matrices. Trivially it is sharp by classic optimization technics on the weights. □

Remark 3.36 (Scope of the Mixture Approximation Guarantee)**.** *Theorem 3.35 provides a constructive sharp upper bound on the information projection of the target mixture onto the GM obtained by applying the information projection to the Gaussian family for each component separately. In particular, each Gaussian component q_j is the unique minimizer of $D_{\mathrm{KL}}(P_j : q)$ over nondegenerate Gaussian PDF q on $\mathbb{R}^D$. The theorem then uses joint convexity of the KL divergence to establish*

$$D_{\mathrm{KL}}\left(\sum_{j=1}^{M} \lambda_j P_j : \sum_{j=1}^{M} \lambda_j q_j\right) \leq \sum_{j=1}^{M} \lambda_j D_{\mathrm{KL}}(P_j : q_j). \tag{3.134}$$

This inequality does not establish that the resulting GM is a global minimizer of the information projection divergence over the family of all GMs. In particular, the componentwise optimality of each q_j does not, by itself, imply global optimality of the mixture, since the mixture divergence depends jointly on all component distributions and their mixture coefficients. Accordingly, the result should be interpreted as a componentwise optimal construction accompanied by a sharp upper bound for the global divergence, rather than as a solution to the global GM optimization problem.

Corollary 3.37 (Uniform Gaussian-Mixture Approximation of an Embedded Simplicial Complex)**.** *Under the assumptions of Theorem 3.35, let $\mathcal{S}$ be a finite, pure, d-dimensional simplicial complex embedded in $\mathbb{R}^D$, with realization*

$$\Omega = |\mathcal{S}|, \quad 0 < \mathrm{Vol}_d(\Omega) < +\infty. \tag{3.135}$$

For every maximal simplex σ_j^d, let

$$Y_j \sim \mathcal{U}_{\sigma_j^d} \tag{3.136}$$

and independently let

$$Z_j \sim \mathcal{N}_{N_j}(z_j|0, \Sigma_{N,j}), \tag{3.137}$$

where $\Sigma_{N,j} : N_j \to N_j$ is positive definite on the orthogonal normal space. Define

$$P_j = \mathcal{L}(Y_j + Z_j) \tag{3.138}$$

and the volume-based weights

$$w_j = \frac{\mathrm{Vol}_d(\sigma_j^d)}{\mathrm{Vol}_d(\Omega)}. \tag{3.139}$$

Consider the Gaussian-smoothed mixture

$$P_\Omega^{\mathrm{sm}} = \sum_{j=1}^{M} w_j P_j \tag{3.140}$$

and its GM representation

$$Q_\Omega = \sum_{j=1}^{M} w_j q_j, \tag{3.141}$$

Preprint

where q_j denotes the unique Gaussian information projection of P_j. Then

$$D_{\mathrm{KL}}\left(P_\Omega^{\mathrm{sm}} : Q_\Omega\right) \leq C_d, \tag{3.142}$$

where

$$C_d = \log\left(\frac{(2\pi e)^{d/2} d!}{(d+1)^{(d+1)/2}(d+2)^{d/2}}\right). \tag{3.143}$$

In particular, the upper bound depends only on the intrinsic dimension d and is independent of the ambient dimension, the geometry and embedding of the simplicial complex, and the normal covariance matrices.

Proof. For every maximal simplex σ_j^d, the uniform distribution on σ_j^d corresponds to the Dirichlet PDF with parameter vector

$$\alpha_j = (1, \ldots, 1) \in \mathbb{R}^{d+1}. \tag{3.144}$$

Consequently,

$$a_j = d+1, \quad h(\mathcal{D}(x_T|1,\ldots,1)) = -\log(d!). \tag{3.145}$$

Substituting these expressions into the component divergence formula of Theorem 3.35 gives

$$C_{d,\alpha_j} = \frac{d}{2}\log(2\pi e) - \frac{d+1}{2}\log(d+1) - \frac{d}{2}\log(d+2) + \log(d!). \tag{3.146}$$

Hence,

$$C_{d,\alpha_j} = C_d \tag{3.147}$$

for every maximal simplex. Furthermore, since distinct maximal simplices intersect only in faces of dimension strictly smaller than d, their pairwise intersections have zero d-dimensional Hausdorff measure. Therefore,

$$\sum_{j=1}^{M} \mathrm{Vol}_d(\sigma_j^d) = \mathrm{Vol}_d(\Omega), \tag{3.148}$$

which implies

$$\sum_{j=1}^{M} w_j = 1. \tag{3.149}$$

Applying Theorem 3.35 with these volume-based weights yields

$$D_{\mathrm{KL}}\left(P_\Omega^{\mathrm{sm}} : Q_\Omega\right) \leq \sum_{j=1}^{M} w_j C_{d,\alpha_j} = C_d \sum_{j=1}^{M} w_j = C_d. \tag{3.150}$$

This establishes the result. □

Remark 3.38 (Uniform PDF Weights)**.** *The measure based weights satisfy*

$$\mathcal{U}_\Omega = \sum_{j=1}^{M} w_j \mathcal{U}_{\sigma_j^d}, \tag{3.151}$$

Preprint

where the equality is understood as an identity of probability measures with respect to d-dimensional Hausdorff measure. The Gaussian-smoothed distribution P_Ω^{sm}, however, is obtained by adding independent Gaussian PDF on the normal spaces of the individual simplices. Thus, the corollary establishes a finite information projection for the Gaussian-smoothed uniform PDF, rather than for the original singular PDF $\mathcal{U}_\Omega$ with respect to ambient Lebesgue measure.

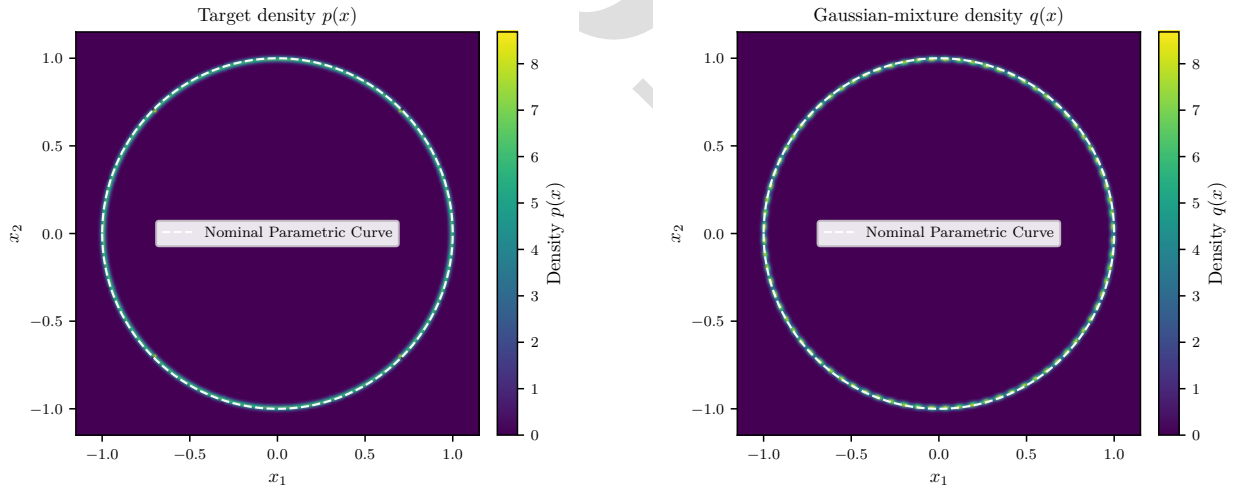
Remark 3.39 (Independence of Normal Covariances)**.** *The bound holds for arbitrary covariance operators $\Sigma_{N,j} : N_j \to N_j$ that are positive definite on their respective normal spaces. In particular, these operators need not be identical across simplices. This independence follows from the orthogonal decomposition into tangent and normal components: the normal Gaussian PDF contribution cancels exactly in each component-wise information projection.*

Remark 3.40 (General Mixture Weights)**.** *Theorem 3.35 holds for arbitrary strictly positive mixture weights. The volume-based weights in Corollary 3.37 are specifically chosen to represent the uniform PDF before adding the Gaussian PDF on the simplex normal space.*

4. An Example

We illustrate the theoretical results and remarks through the GM approximation of a polygonal approximation to a curve in $\mathbb{R}^2$.

Example 4.1 (GM Approximation of a Curve in $\mathbb{R}^2$)**.** *We consider a circular parametric curve embedded in the two-dimensional Euclidean space. The purpose of this example is to illustrate the construction of a GM approximation for a curved geometric object through a finite collection of embedded 1-simplices.*



(A) Target density $p(x)$. (B) Gaussian-mixture density $q(x)$.

Figure 1. Comparison of the target PDF (uniform PDF) and its GM representation for the circular curve which is almost indistinguishable visually.

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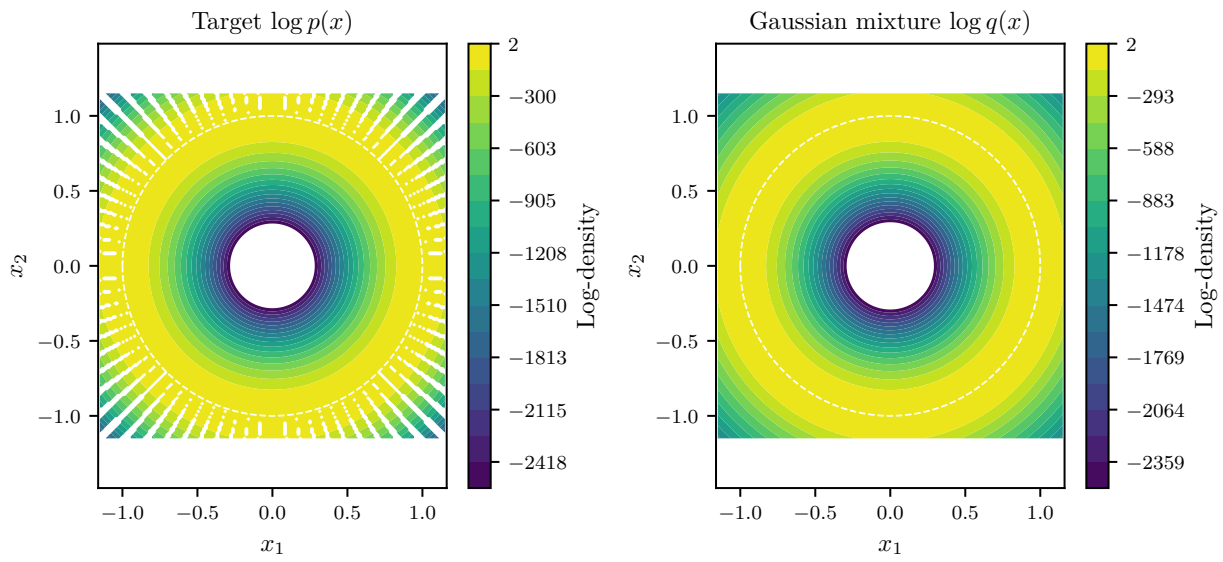


FIGURE 2. Comparison of the target log-density $\log p(x)$ (left) and the Gaussian-mixture log-density $\log q(x)$ (right) for the polygonal approximation of the unit circle. Both distributions are constructed using $M = 80$ equally weighted segment components and normal noise standard deviation $\sigma_{\mathrm{n}} = 0.01$.

Let $\gamma : [-\pi, \pi] \to \mathbb{R}^2$ denote the unit-circle parametrization

$$\gamma(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}. \tag{4.1}$$

The corresponding continuous curve is

$$\Gamma = \{\gamma(t) : t \in [-\pi, \pi]\}.$$

To construct a simplicial approximation, we discretize the parameter interval into M equal subintervals. Specifically, let

$$t_j = -\pi + \frac{2\pi j}{M}, \qquad j = 0, \dots, M, \tag{4.2}$$

and define the corresponding vertices by

$$v_j = \gamma(t_j), \qquad j = 0, \dots, M. \tag{4.3}$$

Since $\gamma(-\pi) = \gamma(\pi)$, we have $v_0 = v_M$. The resulting polygonal approximation consists of the embedded 1-simplices

$$\sigma_j = \mathrm{conv}\{v_{j-1}, v_j\}, \qquad j = 1, \dots, M. \tag{4.4}$$

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For each simplex, define its length, unit tangent, and unit normal by

$$\ell_j = \|v_j - v_{j-1}\|, \tag{4.5}$$

$$t_j = \frac{v_j - v_{j-1}}{\ell_j}, \tag{4.6}$$

$$n_j = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} t_j. \tag{4.7}$$

Because the vertices are equally spaced in the angular parameter, all polygonal segments have the same length,

$$\ell_j = 2\sin\left(\frac{\pi}{M}\right), \qquad j = 1, \ldots, M. \tag{4.8}$$

Figure 3 illustrates the continuous nominal curve and its polygonal approximation.

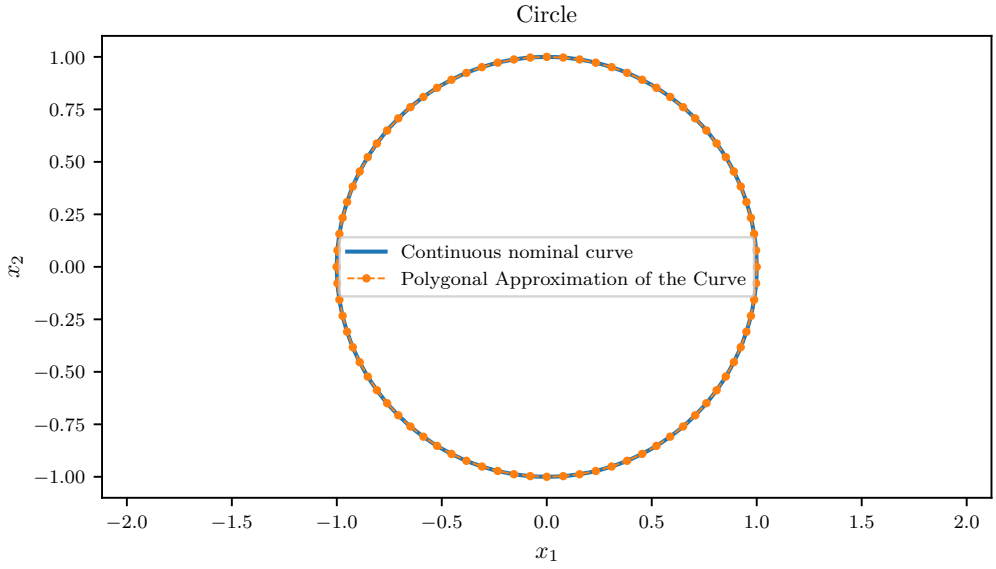


FIGURE 3. Unit-circle parametric curve and its polygonal approximation by $M = 80$ embedded 1-simplices. The blue curve represents the continuous nominal parametric curve, while the orange polygon represents its piecewise-linear approximation.

We associate a probability distribution with each embedded segment by introducing independent random variables

$$U_j \sim \mathcal{U}(u_j|0,1), \quad Z_j \sim \mathcal{N}(z_j|0,1), \quad U_j \perp Z_j. \tag{4.9}$$

The corresponding RV is defined as

$$X_j = v_{j-1} + U_j(v_j - v_{j-1}) + \sigma_{\mathrm{n}} Z_j n_j, \qquad j = 1, \ldots, M, \tag{4.10}$$

where $\sigma_{\mathrm{n}} > 0$ denotes the standard deviation of the Gaussian perturbation in the normal direction.

Thus, the tangential coordinate is uniformly distributed along the segment, whereas the normal coordinate follows a centered Gaussian distribution. The resulting component distribution is absolutely continuous with respect to two-dimensional Lebesgue measure.

Let p_j denote the density of X_j. For $x \in \mathbb{R}^2$, introduce the local coordinates

$$s_j(x) = t_j^\top (x - v_{j-1}), \tag{4.11}$$

$$r_j(x) = n_j^\top (x - v_{j-1}). \tag{4.12}$$

Then the component density is

$$p_j(x) = \frac{1_{[0,\ell_j]}(s_j(x))}{\ell_j \sqrt{2\pi}\sigma_{\mathrm{n}}} \exp\left(-\frac{r_j(x)^2}{2\sigma_{\mathrm{n}}^2}\right). \tag{4.13}$$

We select the segment index according to the arc-length weights

$$w_j = \frac{\ell_j}{\sum_{k=1}^M \ell_k}, \qquad j = 1, \ldots, M. \tag{4.14}$$

Since all segments have equal length in the present construction, these weights simplify to

$$w_j = \frac{1}{M}. \tag{4.15}$$

The resulting target distribution is the mixture

$$p(x) = \sum_{j=1}^M w_j p_j(x). \tag{4.16}$$

For each segment, the mean and covariance of X_j are given by

$$m_j = \mathbb{E}(X_j) = \frac{v_{j-1} + v_j}{2}, \quad \Sigma_j = \mathbb{V}(X_j) = \frac{\ell_j^2}{12} t_j t_j^\top + \sigma_{\mathrm{n}}^2 n_j n_j^\top. \tag{4.17}$$

By the componentwise information-projection result established in this work, the unique forward-KL optimal nondegenerate Gaussian approximation is

$$q_j = \mathcal{N}(x | m_j, \Sigma_j). \tag{4.18}$$

In particular, the Gaussian covariance captures the uniform variation along the segment and the independent Gaussian perturbation in its normal direction.

For the uniform distribution on a 1-simplex, the minimum componentwise divergence is

$$D_{\mathrm{KL}}(p_j : q_j) = C_1 = \frac{1}{2} \log\left(\frac{\pi e}{6}\right) \approx 0.17649 \quad \textit{nats.} \tag{4.19}$$

This quantity is independent of the segment length, orientation, location, and normal noise variance.

We construct the Gaussian-mixture approximation using the same component weights:

$$q(x) = \sum_{j=1}^M w_j q_j(x). \tag{4.20}$$

By joint convexity of the Kullback–Leibler divergence, we obtain

$$D_{\mathrm{KL}}(p : q) \le \sum_{j=1}^M w_j D_{\mathrm{KL}}(p_j : q_j) = \sum_{j=1}^M w_j C_1 = C_1 = \frac{1}{2} \log\left(\frac{\pi e}{6}\right) \approx 0.17649 \quad \textit{nats.}$$

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Consequently, the information projection divergence between the target distribution and its GM divergence admits a bound that is independent of the number of segments, the polygonal geometry, and the normal noise variance.

The target log-density $\log p(x)$ *and the Gaussian-mixture log-density* $\log q(x)$ *are evaluated and compared over the ambient two-dimensional space. Figure 2 shows the resulting log-density representations. The left panel displays the target distribution, whereas the right panel displays its Gaussian-mixture approximation.*

For the Monte Carlo simulation we estimate the KL divergence $D_{\mathrm{KL}}(p : q)$ *using* $\#Samples = 30000$*. For* $\#Segments = 80$*,* $\sigma_{\mathrm{n}} = 0.01$*, and random seed* 42*, the estimated divergence is*

$$\widehat{D}_{\mathrm{KL}}(p : q) = 0.045109, \tag{4.21}$$

with standard error 2.987×10^{-4}*. The corresponding* 95% *confidence interval is*

$$[0.044524, 0.045695]. \tag{4.22}$$

This estimate lies strictly below the theoretical upper bound

$$D_{\mathrm{KL}}(p : q) \leq \frac{1}{2} \log\left(\frac{\pi e}{6}\right) \approx 0.17649. \tag{4.23}$$

Thus, the numerical results are consistent with the theoretical mixture-divergence guarantee and indicate a substantially smaller empirical divergence than the established upper bound.

The comparison illustrates how the individual segment-wise Gaussian approximations combine to produce a full-dimensional GM around the polygonal approximation of the continuous circular curve. The resulting distribution retains the overall geometric structure of the target while replacing the individual segment distributions by nondegenerate Gaussian components.

Remark 4.2 (General Curves)**.** *Although the present example focuses on a circular parametric curve, the underlying construction is by no means restricted to this particular geometry. It applies to general curves through their polygonal approximations, with each segment inducing a probabilistic geometric primitive and its corresponding Gaussian component. In particular, the framework accommodates smooth and closed curves, open curves, non-regular curves, as well as self-intersecting geometries and unions of curves.*

This generality has already been demonstrated from a computational perspective in the complementary work [4, 5], where a broad collection of canonical plane curves is employed as test cases to investigate the expressive capabilities of geometric GM representations. These experiments demonstrate that the resulting GM faithfully capture the local tangential and normal structure, as well as the global geometry of the underlying curves.

The present work complements this computational investigation by providing the corresponding mathematical foundations and theoretical guarantees for the Gaussian approximation. In this sense, the circular example considered here serves as a representative instance of a substantially more general geometric probabilistic construction, rather than as a geometry-specific application.

Remark 4.3. *The example illustrates the application of the componentwise Gaussian information projection and the global GM divergence bound to a closed parametric curve. Although the underlying continuous curve is nonlinear, its polygonal approximation admits a direct representation through embedded* 1*-simplices.*

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Increasing the number of segments improves the geometric resolution of the polygonal approximation. Nevertheless, the theoretical information projection upper bound remains unchanged because the componentwise divergence depends only on the intrinsic dimension $d = 1$.

Furthermore, changing the normal PDF standard deviation modifies the spatial spread of the target and GM components without affecting the theoretical bound.

Remark 4.4 (Connection to Embedded Simplicial Complexes)**.** *The polygonal approximation constructed above is a one-dimensional geometric simplicial complex embedded in $\mathbb{R}^2$. In particular, its maximal simplices are precisely the line segments $\{\sigma_j\}_{j=1}^{M}$.*

This provides a concrete instance of the embedded simplicial complexes introduced in Definition 3.7. Moreover, the segments intersect only at common vertices, so the intersection property of Proposition 3.10 is satisfied. Consequently, the polygonal realization admits the decomposition into maximal simplices required by the GM theoretical upper bound construction.

Thus, the example illustrates how a nonlinear geometric object can be represented through a finite embedded simplicial complex, allowing the theoretical results to be applied directly.

Remark 4.5 (Uniform Distributions and Dirichlet Distributions)**.** *The uniform distribution along each line segment corresponds to the Dirichlet distribution with parameter vector $\alpha = (1, 1)$. Indeed, the barycentric coordinates of a point on a 1-simplex are distributed according to $\mathcal{D}(x|1, 1)$, which is simply the uniform distribution on the standard 1-simplex.*

The component distributions in (4.10) *are therefore special cases of the Dirichlet–Gaussian construction introduced in Definition 3.26.*

Furthermore, the component means and covariances in Equation (4.17) *are precisely the uniform-simplex moment formulas established in Proposition 3.23.*

Hence, the numerical construction is not specific to the circular geometry but is an instance of the general Dirichlet–Gaussian framework developed in the preceding sections.

Remark 4.6 (Componentwise Gaussian Information Projection)**.** *The Gaussian components in* (4.18) *are not chosen heuristically. They are the unique information projection of the corresponding distributions onto the Gaussian family, as established by Theorem 3.30.*

In particular, the Gaussian means and covariance matrices are determined entirely by moment matching. The covariance decomposition

$$\Sigma_j = \underbrace{\frac{\ell_j^2}{12} t_j t_j^\top}_{\text{tangential variation}} + \underbrace{\sigma_n^2 n_j n_j^\top}_{\text{normal perturbation}} \tag{4.24}$$

makes the geometric interpretation of the information projection explicit.

The first term captures the uniform distribution along the segment, whereas the second preserves the Gaussian PDF introduced in the normal direction. This example therefore illustrates how the abstract information projection result translates into a geometrically interpretable Gaussian approximation.

Remark 4.7 (Normal Gaussian Independence)**.** *An important consequence of the componentwise information projection formula is that the optimal approximation error does not depend on the normal PDF variance σ_{n}^2.*

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Although changing σ_{n} modifies the spatial spread of both the target components and their information projection onto the Gaussian family, the normal Gaussian contribution cancels exactly in the componentwise information projection. Consequently,

$$D_{\mathrm{KL}}(p_j : q_j) = \frac{1}{2}\log\left(\frac{\pi e}{6}\right) \approx 0.17649 \tag{4.25}$$

for every segment and every $\sigma_{\mathrm{n}} > 0$.

This illustrates the general independence of the normal covariance operators established in Remark 3.17 and shows that the KL divergence is governed by the intrinsic Dirichlet–Gaussian distribution on the embedded simplex rather than the ambient Gaussian PDF.

Remark 4.8 (Global Mixture Guarantee)**.** *The construction of the full GM in Equation* (4.20) *illustrates the passage from componentwise information projections to a GM with a sharp upper bound.*

Since the GM weights are identical for the target and approximating distributions, joint convexity of the KL divergence, as established in Proposition 3.34, gives the bound in Theorem 3.35. For the present example, the intrinsic dimension is $d = 1$, and the resulting upper bound is

$$D_{\mathrm{KL}}(p : q) \leq C_1 = \frac{1}{2}\log\left(\frac{\pi e}{6}\right) \approx 0.17649. \tag{4.26}$$

Importantly, this guarantee is independent of the number of segments M, their lengths, their orientations, and the normal PDF variance. Thus, increasing the number of segments improves the geometric resolution of the polygonal approximation without changing the theoretical sharp upper bound on the GM divergence. As emphasized in Remark 3.36, the result establishes a sharp upper bound for the constructed GM, rather than global optimality over all possible GMs.

5. Conclusion

We derived the unique Gaussian information projection of Dirichlet–Gaussian PDFs on embedded simplices, obtaining a closed-form minimum information projection which is invariant up to an affine isomorphism and independent of the ambient dimension and normal covariance. By joint convexity of the KL divergence, we further established the sharp intrinsic-dimension-dependent KL divergence upper bounds for finite GMs, giving sharp approximation guarantees for GMs associated with embedded simplicial complexes. A numerical example on a circular curve illustrates the theoretical results and their applicability to embedded manifolds represented by simplicial complexes.

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Ali Darijani[⊛□#¶⊥]
ORCID: https://orcid.org/0009-0000-4659-093X
Research Staff, Fraunhofer IOSB of Fraunhofer-Gesellschaft, Germany
Research Staff, KIT Department of Computer Science, Germany
Alumnus, RWTH Aachen University, Germany
Alumnus, Sharif University of Technology, Iran
Email address: ali.darijani@iosb.fraunhofer.de
Email address: ali.darijani@kit.edu
Email address: ali.darijani@rwth-aachen.de
Email address: ali.darijani@alum.sharif.edu

Emadedin Soufian[□#¶⊥]
ORCID: https://orcid.org/0009-0006-5777-4623
Research Staff, Det Norske Veritas (DNV), Germany
Alumnus, RWTH Aachen University, Germany
Alumnus, Iran University of Science and Technology, Iran
Email address: emadedin.soufian@dnv.com
Email address: emadedin.soufian@rwth-aachen.de

Jürgen Beyerer[⊛⊥]
ORCID: https://orcid.org/0000-0003-3556-7181
Director, Fraunhofer IOSB of Fraunhofer-Gesellschaft, Germany
Professor, KIT Department of Computer Science, Germany
Alumnus, Karlsruhe Institute of Technology, Germany
Email address: juergen.beyerer@iosb.fraunhofer.de
Email address: juergen.beyerer@kit.edu

Symbol Legend
⊛: Idea, □: Math, #: Programming, ¶: Writing, ⊥: Critique

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